GAVEL

Every scam gets a fair trial.

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The problem

Scams are not won in moments of ignorance. They are won in moments of decision. A parcel-fee SMS, a bank alert, a "job offer" — the person at the centre has seconds to act, and every second is engineered against them. Today's defences answer the wrong question: "is this domain on a blocklist?" They return a score and call it a verdict, they never show their working, and they can never say *"this one is probably genuine"* — even though honest bills, delivery notices and bank alerts arrive every single day.

GAVEL is built for the people at the centre of that decision: everyday mobile users and the families they protect — in Malaysia and anywhere a scam arrives by phone.

The idea

GAVEL turns scam defence into a courtroom. You are the judge. Anything suspicious — a message, a link, a QR code, a document, a voice note — is booked as a case and put on trial.

- **The Prosecutor** opens the case and reads the strongest exhibits: a calibrated text classifier trained on labelled scam corpora, offline URL anatomy, QR decoding, document extraction. Those tools run deterministically before any officer is drafted — the models cannot choose what to inspect. Every finding is filed as a numbered **exhibit**, pinned to the exact words in your message that raised it.
- **The Defence** argues the benign reading — the brand's real domain, the shape of a genuine transactional notice, the message that *warns* about scams instead of running one.
- **The Clerk** writes the bench memo: the charge, the evidence, the counter-evidence, and a recommended finding with an honest confidence band.
- **You** swing the gavel: **GUILTY**, **ACQUITTED**, or **HUNG JURY**. When the evidence genuinely cannot support either reading, the court says so — it refuses to dress uncertainty up as a decision.

The verdict is the beginning, not the end. A guilty verdict unlocks a **protection order**: what to do in the next five minutes, a cooling-off timer, a redacted report ready to file, one-tap alerts to the people who trust you. An acquittal protects the delivery notice that really is from the courier.

The four fronts, one product

DIRECTION	IN GAVEL
Phishing detection	The trial itself. Exhibits are quoted back onto your own message, and the full **court transcript** records what each agent checked, which tool it called, and why — explainability as evidence, not as a promise.
Scam prevention	The evidence chamber. Link and QR anatomy before anything is clicked: homoglyphs, subdomain chains, credentials-in-URL, shorteners, and where a decoded QR actually goes.
Security learning	Law School. Scam types written up as case law distilled from official guidance, a **Daily Docket** (one case to judge a day), drills, and ranks earned by judging real cases.
Safer everyday decisions	The sentencing desk. Next-five-minutes checklists, a cooling-off timer for big transfers, redacted family alerts, and report-ready summaries.

Agentic by design, safe by construction

GAVEL runs an orchestration graph: intake → deterministic evidence pass (works with zero network) → three drafted officers (a prosecutor, a defence and a clerk, as language-model agents) → the judge's gavel.

The suspicious message is treated as untrusted **evidence, never instructions** — attempts to steer the agents are docketed as *witness tampering*. The tools are deliberately not agent-selected: a model that chooses which evidence to look at can be steered by the evidence itself.

Two mechanisms make the court trustworthy in a way a single chatbot can never be:

- **The verification gate.** Every drafted speech is checked before it may speak: real prose structure, and **every word inside quotation marks must exist character-for-character in the court record**. A draft that invents evidence is discarded. Fabricated quotes reach the screen zero times, by construction rather than by promise.
- **The scripted bench.** When no model service is available — offline, rate-limited, or failing the gate — the officers' arguments are compiled deterministically from the exhibits themselves. The courtroom completes every trial, with or without a network. The product cannot fail in front of a user, a judge, or a demo audience.

Data and measured rigor

- Text detector trained and calibrated on the **UCI SMS Spam Collection** (5,574 labelled messages): held-out AUC **0.9990**, calibration error **0.0060**.
- The complete shipped engine, evaluated on 1,032 held-out messages: **precision 96.8%**, scam recall **96.0%**, false alarms on genuine messages **0.44%**, sent to hung jury **0.87%**, **0.06 ms** per case on device.
- A 24-case authored golden set (modern parcels, tax lures, genuine notices, two designed to be undecidable) passes **24/24** in the shipped test suite.
- One engine, two implementations: the Python training pipeline and the live TypeScript runtime are parity-tested, and the published numbers are regenerated by the repository's own evaluation scripts.

Limitations and scope

GAVEL states its limits as plainly as its claims. The text classifier is trained on English-labelled SMS corpora; local slang beyond that leans on pattern rules. Links are judged by anatomy, not live reputation. Without a model service, the officers' speeches are compiled deterministically — the trial completes, the prose is plainer. GAVEL blocks nothing and contacts no one; the decision stays with the person.

What ships is complete: the deployed app, the measured engine, a demo film, and the test suite that regenerates every published number. Docketed next: a consent-based public-record check, Bahasa Malaysia and regional corpora, and case law that grows from real verdicts.

Business case

- **Who pays.** The family buyer — the person who forwards the parcel SMS to the group chat — at RM4.90/month; and the bank, telco or e-wallet that would rather prevent a transfer than reverse one, at a pilot licence.
- **Market (bottom-up; assumptions, not forecasts).** 8.0M Malaysian households × RM4.90/month ≈ **RM470M/year** consumer TAM; 27 banks, 5 telcos and 10 e-wallets × RM250k pilot ≈ **RM10M/year** initial institutional segment.
- **Why it holds.** Local-first forensics keep marginal cost near zero — tokens are paid only when the court speaks — and one engine serves both the consumer app and an institutional API.

Impact and viability

Scam harm keeps accelerating; awareness programmes lecture, and detection tools emit scores — neither puts the person in charge of the decision. GAVEL is human-centred by construction: the AI investigates, the person decides, and every case teaches the next one. It installs nothing, runs on a free tier, and protects privacy — personal identifiers are masked before anything is shared, and in local mode the message never leaves the device. The product ships with both global scam patterns and local ones (parcel-fee SMS, e-wallet traps, government-impersonation) so it is useful from the first day in the hands of the people who need it most.

GAVEL — because the best scam detector is a person who has learned to judge.

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